

Zebrafish DCF ROS Normalization Methods

Within-session normalization, outlier handling, and nested statistical analysis

zebrafish-ros v1.0.0 · github.com/ebalderasr/zebrafish-ros-normalization-online

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Methods

Manuscript-ready description of the data processing, normalization, and statistical analysis implemented in this repository. Three versions are provided: an extended version for the Methods section of a research article, a compact version for short-format journals, and a single paragraph for a strict word limit.

The analysis mirrors the one in hyper-normalizer, which handles the ratiometric HyPer measurements of the same study. Both assays are normalized against the control acquired in the same session and analysed with the session as the unit of replication, so the two sets of results can be reported side by side. The differences specific to DCF, namely the median anchor and the outlier branch, are stated where they arise.

Text in square brackets marks acquisition details that the code does not determine and that the authors must supply. Sample sizes refer to the dataset processed at the time of writing and should be checked against the final dataset.

Extended version

Fluorescence quantification and data structure

Intracellular reactive oxygen species were measured with the fluorogenic probe 2',7'-dichlorodihydrofluorescein diacetate (H₂DCFDA), which yields the fluorescent product DCF upon oxidation. For each embryo, a single fluorescence intensity was quantified [*add probe concentration and incubation time, microscope, objective, excitation and emission settings, and region-of-interest definition*].

Unlike a ratiometric sensor, DCF reports a single channel, so its absolute intensity carries no internal correction for probe loading, embryo size, or illumination. Values are therefore comparable only among embryos imaged in the same session, which is what makes the within-session anchor described below necessary rather than merely convenient.

The individual embryo was the unit of measurement. The independent acquisition session,

defined as a cohort of embryos stained, treated, and imaged on the same day, was the unit of replication. Values were exported as one matrix per experimental group and per pharmacological panel, with rows corresponding to individual embryos and columns to conditions, and converted to long format in which each record comprises the group, acquisition date, condition, and intensity of a single embryo. Rows are not paired: two values on the same row are two different embryos imaged on the same date, not one embryo measured twice. Not every compound was tested in every session, so absent measurements were missing by design; they were excluded and not imputed. Where one group was split across panels, the shared vehicle control was recorded in each panel’s sheet and duplicated records were collapsed to a single series per group and session before analysis. The final dataset comprised [568] embryos across [2] experimental groups and [7] acquisition sessions, with each compound tested in [4 to 5] sessions.

Outlier handling

Within each group, acquisition date, and condition, embryos falling outside Tukey’s bounds were flagged:

$$L_{\text{low}} = Q_1 - 1.5 \text{ IQR}, \quad L_{\text{high}} = Q_3 + 1.5 \text{ IQR}, \quad \text{IQR} = Q_3 - Q_1.$$

Groups of fewer than four embryos were left unflagged, since their quartiles are interpolated from too few points for the rule to be informative. Two parallel analyses were carried out and are reported in full: one retaining every embryo and one excluding the flagged ones. Because removing an embryo from a control group changes that session’s anchor, the normalization was recomputed from the filtered data rather than applied first and filtered afterwards. [*State which branch the reported figures use, and confirm that the conclusions hold in both.*] In the present dataset [28] of [568] embryos ([4.9] %) were flagged.

Within-session normalization

Absolute DCF intensity varied between acquisition sessions as a consequence of probe loading, illumination power, detector gain, and developmental stage. Across the present dataset the session median of the vehicle control varied by up to [2.6]-fold, which exceeds the treatment effects of interest. Uncorrected, this variation would dominate the between-group comparison. The variation is multiplicative, and the raw intensity of embryo i was modelled as

$$I_i = \mu_t \cdot \beta_{g,d} \cdot \varepsilon_i,$$

where μ_t denotes the effect of condition t , $\beta_{g,d}$ a session-specific gain factor for group g on date d , and ε_i the residual between-embryo variability. Each measurement was expressed relative to the median of the vehicle (DMSO) controls acquired in the same group and the same session,

$$\text{ratio}_i = \frac{I_i}{\tilde{I}_{g,d}^{\text{veh}}}, \quad \tilde{I}_{g,d}^{\text{veh}} = \text{median} \left(I_j : j \in \text{veh}(g, d) \right),$$

and reported on a log base 2 scale as $\log_2 \text{FC}_i$. The median was preferred over the mean because single-channel intensities carry occasional extreme values; the mean gives the same conclusions and is available as an option in the software. The denominator was always the concurrent control of the same group and session. No pooled or global control mean was used, and no anchor was shared between groups.

This transformation is invariant to any rescaling of a complete session, so session-level differences in gain are eliminated rather than reduced. Its effect was verified directly: the coefficient of variation of the daily medians across sessions fell from [0.29 to 0.45] before normalization to [0.04 to 0.15] after it, for every condition. Sessions with fewer than three control embryos were flagged as having a poorly determined anchor.

Two consequences constrained the subsequent analysis. All embryos from a given session share the same denominator and are therefore not mutually independent. The vehicle group is also centred on 1 in every session by construction and cannot be treated as a freely varying sample.

Statistical analysis

All analyses were performed on $\log_2$ -transformed values. The logarithm is the appropriate scale for a strictly positive multiplicative quantity. It renders the model above additive, makes symmetric intervals meaningful, since a doubling and a halving are then equidistant from zero, and removes the upward bias affecting the arithmetic mean of ratios.

The independent acquisition session was used as the unit of replication, which avoids pseudoreplication. For each group g , condition t , and session d , the within-session effect was computed as the difference between the mean $\log_2$ intensity of the treated embryos and that of the vehicle controls acquired in the same session,

$$\delta_{t,d} = \overline{\log_2 I}|_{t,d} - \overline{\log_2 I}|_{\text{veh},d}.$$

This difference is taken within a session and on the raw intensities, so $\log_2 \beta_{g,d}$ cancels algebraically. The estimate therefore depends neither on the normalization step, which serves presentation, nor on whether the median or the mean was used as the anchor. The resulting values, one per session, were tested against zero by a two-tailed one-sample t test, with n equal to the number of independent sessions in which the condition and its concurrent vehicle control were both acquired ($n = [4 \text{ to } 5]$ per compound). Effects are reported as the geometric fold change 2^{δ} with its 95 % confidence interval, and p values were adjusted for multiple comparisons within each experimental group using the Holm-Bonferroni procedure.

Each session carries equal weight in this test, independently of how many embryos it contains. As a sensitivity analysis, the test was repeated after excluding sessions in which fewer than three embryos were measured [*state whether the conclusions were unchanged*]. An equivalent linear mixed-effects model, $\log_2 I \sim \text{condition} + (1|\text{session})$, was also fitted for each group with session as a random intercept, estimating the session and condition effects jointly rather than normalizing before testing. Tests treating individual embryos as independent replicates were not used. They disregard the nested structure of the design and yield anti-conservative p values, which for the present dataset were optimistic by up to thirteen orders of magnitude.

Data presentation

Results are displayed as SuperPlots (Lord et al., 2020). Individual embryo values are plotted as small background points, and the mean of each independent session is superimposed as a large colour-coded marker, so that the number of independent replicates and the between-session consistency of each effect are visible. Boxes indicate the median and interquartile range of the individual embryo values, and the dashed line marks the normalized vehicle control at 1. A companion figure plots the raw control anchor of each session, which is the between-session drift that the normalization removes.

Software and data availability

The analysis is implemented twice from a single specification: as a browser-based application that runs entirely client-side through Pyodide, with no data leaving the user’s machine, and as an offline Python package for scripted and reproducible use. Both are available at <https://github.com/ebalderasr/zebrafish-ros-normalization-online> (version 1.0.0, MIT licence), and the application is deployed at <https://ebalderasr.github.io/zebrafish-ros-normalization-online/>. The offline package uses Python 3.10 with NumPy, SciPy, pandas, Matplotlib, seaborn, and statsmodels. An automated test suite verifies that the two implementations produce identical per-embryo $\log_2$ FC values, control anchors, outlier flags, and per-session tables, and checks the defining invariants of the normalization, including invariance to session-level rescaling.

Reference. Lord SJ, Velle KB, Mullins RD, Fritz-Laylin LK. SuperPlots: Communicating reproducibility and variability in cell biology. *J Cell Biol.* 2020;219(6):e202001064. doi:10.1083/jcb.202001064

Compact version

Intracellular reactive oxygen species were measured with H₂DCFDA, and one DCF fluorescence intensity was quantified per embryo. Within each group, acquisition date, and condition, embryos outside Tukey’s 1.5 IQR bounds were flagged, and the analysis was carried out both retaining and excluding them. Because single-channel DCF intensity carries no internal correction and drifts between acquisition sessions, each measurement was normalized to the median of the vehicle (DMSO) controls acquired in the same group and the same session, yielding a $\log_2$ fold change relative to vehicle. The anchor was recomputed after outlier removal, since dropping a control embryo changes that session’s denominator.

Statistical analysis used the independent acquisition session, rather than the individual embryo, as the unit of replication. For each session, the difference between the mean $\log_2$ intensity of treated embryos and that of the concurrent vehicle controls was computed; these session-level differences were tested against zero by a two-tailed one-sample *t* test ($n = [4 \text{ to } 5]$ independent sessions per compound) with Holm-Bonferroni correction within each group. Effects are reported as geometric fold changes with 95 % confidence intervals and displayed as SuperPlots showing per-session means superimposed on individual embryo values. Analyses were performed in Python, in a browser application and an equivalent offline package whose agreement is covered by automated tests; both are available at <https://github.com/ebalderasr/zebrafish-ros-normalization-online>.

Single-paragraph version

For contexts with a strict word limit.

DCF fluorescence intensity was quantified per embryo, embryos outside Tukey’s 1.5 IQR bounds within their own group, date and condition were flagged, and each measurement was normalized to the median of the vehicle (DMSO) controls acquired in the same group and the same acquisition session, which removes the multiplicative session-to-session drift in absolute signal and yields a $\log_2$ fold change relative to vehicle. Inference used the independent acquisition session as the unit of replication: per-session differences between treated embryos and concurrent vehicle controls were tested against zero by a two-tailed one-sample *t* test ($n = [4 \text{ to } 5]$ sessions per compound) with Holm-Bonferroni correction within each group, and effects are

reported as geometric fold changes with 95 % confidence intervals. Analysis code is available at <https://github.com/ebalderasr/zebrafish-ros-normalization-online>.